Full System Architecture, End-to-End Workflow & Codebase Mapping

Comprehensive guide covering authentication, ticket lifecycle, SLA escalation, MySQL database schema, and source file locations.

Executive Summary: The Test Grievance Redressal System is a full-stack, enterprise-grade web application engineered for Test manufacturing plants (H.O, 1F Manesar, 2F Tapukara, 3F Narsapura, 4F Vithalapur). It provides plant-level routing, tamper-proof location binding, multi-stage investigation workflows, automated SLA tracking, Head Office escalations, and interactive analytical reporting.

1. Technical Stack & System Architecture

The system employs a high-performance 3-tier architecture with a decoupled React SPA frontend, an Express.js REST API server, and a MySQL relational database.

Tier / Layer	Technology	Key Role & Responsibility
Frontend Client	React 18 + Vite + Tailwind CSS	Single Page Application (SPA), responsive UI, role-based dashboards, interactive charts & forms.
Backend API	Node.js + Express.js	REST API endpoints, session handling, business logic, SLA evaluation, validation, security logging.
Database Layer	MySQL (mysql2 pool)	Relational storage for 40+ employees/admins, grievances, audit logs, comments, and OTP credentials.
Build & Bundler	Vite 5	High-speed asset compilation, tree-shaking, production builds output directly to client/dist.

2. Complete End-to-End System Workflow

The grievance lifecycle follows a structured 7-step enterprise path from submission to closure:

Step 1: Role-Based Authentication
Associates and Admins log in using their Employee ID / Email and Password on the Login Page. The system verifies credentials against the MySQL database and establishes an authenticated session with user profile and assigned plant location.

Step 2: Auto-Detected Grievance Submission
When an associate lodges a grievance, the system automatically detects their registered plant location (H.O, 1F, 2F, 3F, 4F) and locks it. Manual plant tampering is prevented on both frontend and backend. The associate specifies category, area, priority, description, attachment, and optional anonymity.

Step 3: Sequential Ticket ID & Plant Routing
Upon submission, the backend automatically generates a sequential ticket code (e.g. TEST-1F-2026-008), auto-assigns the ticket to the designated Plant Associate Admin, and creates the first timeline audit log.

Step 4: Admin Investigation & Status Progression
The assigned Plant Admin opens the ticket in the Admin Dashboard to review evidence, write investigation notes, post public or internal comments, and update status through: Submitted -> Under Review -> In Progress -> Resolved -> Closed.

Step 5: Automated SLA Tracking & H.O Escalation
Each ticket is governed by priority SLAs (Urgent: 24h, High: 48h, Normal: 5 days). If an unresolved ticket breaches its SLA or requires executive intervention, it is escalated to Head Office (H.O) Global Admins.

Step 6: Resolution, Feedback & Audit Trail
When resolved, the associate receives resolution remarks, can review the complete chronological action timeline, and submit feedback. All actions are immutably recorded in activity_logs.

Step 7: Analytics & Employee Directory
Executive admins monitor plant-wise performance, average turnaround time (TAT), category distributions, and search the plant associate directory.

3. Database Schema & Storage Architecture

All data is persistently stored in a relational **MySQL Database** named test_grievance_db running on port 3306. The server utilizes pooled connections for high concurrency and zero memory leaks.

Table Name	Description	Key Columns & Types
`users`	User accounts (Associates, Plant Admins, HO Global Admins)	id (PK), emp_id (UNIQUE), name, email (UNIQUE), password_hash, role ('associate'|'admin'), location ('H.O'|'1F'..'4F'), department, designation, phone, is_global_admin
`grievances`	Official grievance tickets with status and routing info	id (PK), ticket_number (UNIQUE), title, category, location, specific_area, priority ('low'|'medium'|'high'|'urgent'), status ('submitted'|'under_review'|'in_progress'|'resolved'|'closed'), description, attachment_url, is_anonymous, escalated_to_ho, submitted_by_id (FK), assigned_to_id (FK), resolution_notes, created_at, updated_at
`comments`	Conversation thread and internal admin investigation notes	id (PK), grievance_id (FK), author_id (FK), message, is_internal (0|1), created_at
`timeline_events`	Chronological audit log of ticket status transitions	id (PK), grievance_id (FK), type, description, performed_by_id (FK), created_at
`activity_logs`	System-wide security and access audit trail	id (PK), user_id (FK), action, details, ip_address, created_at
`otp_codes`	Time-bound verification tokens for self-service password reset	id (PK), emp_id, code, email, expires_at

4. Source Code & File Location Reference

Direct mapping of where every feature and function is implemented in the repository:

File Path	Layer	Key Features & Functions Written in this File
`client/src/pages/LoginPage.jsx`	Frontend	Employee/Admin login interface, credential input validation, and **Download Credentials Button** for the 40 test accounts.
`client/src/pages/SubmitGrievanceModal.jsx`	Frontend	Grievance submission form with **auto-detected plant location display** (no manual selector), file attachment upload, and priority flags.
`client/src/pages/EmployeeDashboard.jsx`	Frontend	Associate workspace showing lodged grievances, real-time status trackers, SLA badges, and quick submission action button.
`client/src/pages/AdminDashboard.jsx`	Frontend	Admin workbench with plant/status/priority filters, search, quick stats cards, and ticket management actions.
`client/src/pages/GrievanceDetailModal.jsx`	Frontend	Ticket investigation view, status updater dropdown, internal notes toggle, conversation chat, and timeline audit logs.
`client/src/pages/AnalyticsPage.jsx`	Frontend	Interactive metrics dashboard: SLA resolution rate, plant-by-plant grievance breakdown, category distributions.
`client/src/pages/DirectoryPage.jsx`	Frontend	Plant-wise Associate & Administrator directory with instant search and contact information cards.
`client/src/pages/ForgotPasswordPage.jsx`	Frontend	3-Step OTP password reset workflow: Request OTP -> Verify OTP code -> Set new password.
`server/server.js`	Backend	Express REST API server: Authentication endpoints (/api/auth/*), grievance CRUD (/api/grievances/*), analytics (/api/analytics/*), CORS & static file hosting.
`server/db.js`	Backend / DB	MySQL connection pool, automatic table schema initialization (initDB()), demo seed data generator (40 users + grievances), and all SQL query handlers (createGrievance(), findUserByCredentials(), updateGrievance(), etc.).

`server/.env`	Config	Database configuration environment variables (DB_HOST, DB_USER, DB_PASSWORD, DB_NAME, PORT).

Note on Data Security: Passwords in production are salted and hashed. Plant boundaries are strictly enforced so that Plant Admins only access grievances belonging to their facility, while HO Admins maintain global oversight.